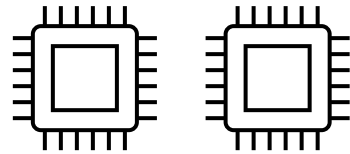


EMS Project



Fingertip-Hand ECG and Heartbeat Meter System

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1- Introduction

Short Description



During a gym/physical exercise, for example in a static bike, the indication of the heartbeat rate and visualization of the user *electrocardiogram* (ECG) can be useful to optimize the training plan. This would be even more useful if the acquired data is wirelessly transmitted (e.g., WiFi) to a smartphone or to a database and processing server. The ECG signal sensor (simple contact electrodes made of metal) can be incorporated into the bicycle's handlebars, as shown in the figure.

The Cardiac muscle (heart) is able to maintain a periodic movement determined by specific controlling neurons. These cells emit periodic electrical pulses which are processed by the muscle cells originating a repetitive cardiac movement. Interestingly these body bioelectrical controlling signals propagate up to the skin surface, meaning that it is possible to detect them using appropriate contact sensors (electrodes). The typical acquired electrical signal is shown in figure 1. It can be observed the existence of a repetitive “strong” pulse, known as the QRS Complex segment. By detecting this QRS segment one can determine the heartbeat rate.

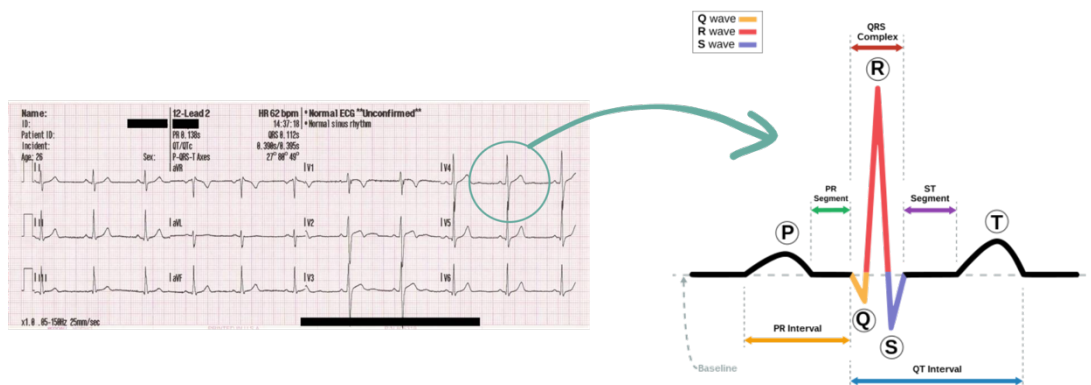


Figure 1 – Typical electrocardiogram (<https://en.wikipedia.org/wiki/Electrocardiography>).

An introductory example can be found in the application note SLAA280B, “Heart-Rate and EKG Monitor Using the MSP430FG439” from Texas Instruments. <https://www.ti.com/lit/pdf/slaa280>

Requirements and system architecture

The current project option consists of designing, implementing, and testing a fingertip ECG and hearth beat acquisition and visualization IoT system.

The captured raw and/or processed sensor data is transmitted, via USB, to a computer where extra processing can be done and visualization using an easy-to-use graphical user interface (GUI). Additionally, it is also required that the same data be transmitted wirelessly to a processing server running an IoT application.

List of minimum requirements:

- Determine and show the heartbeat rate value in real-time;
- The system should show an ECG graphic (electrocardiogram);
- The IoT node system should be able to operate with a 9V battery supply.
- Sensor data should be sent to a computer/server using a USB serial port and a WiFi wireless connection.
- In the computer, the values of the heartbeat rate and an ECG graphic should be presented in a graphical user interface (GUI) window;

This list is not complete. Please complete it before starting the design phase of the project.

A simplified block diagram of the system is depicted in figure 2.

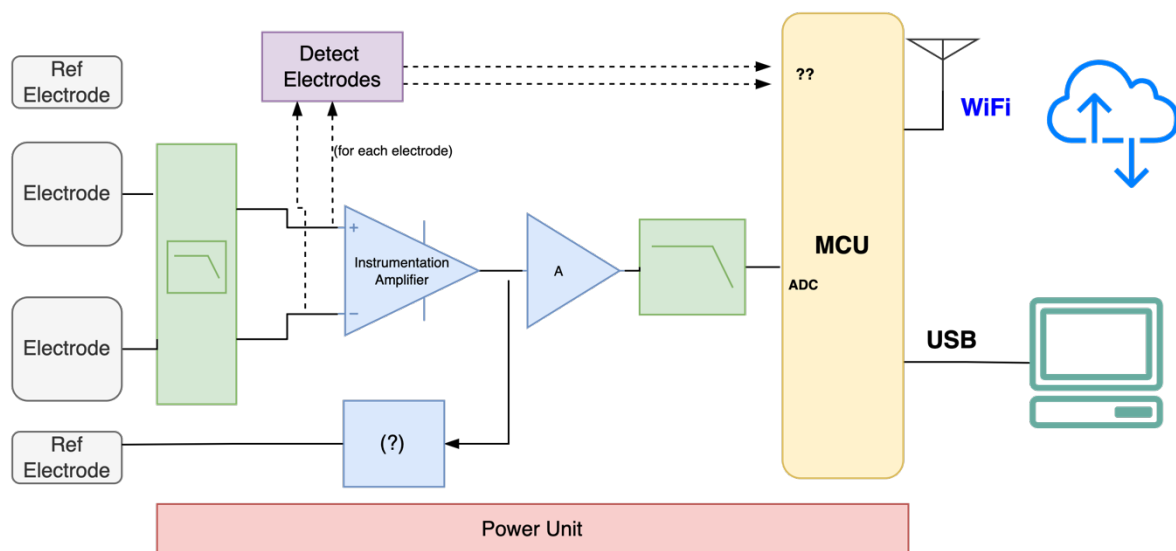


Figure 2 - Simplified block diagram of the SpO2 meter.

2- Hardware and Software/Firmware Considerations

2.1- AFE

The analog front-end is composed by a set of skin contact (dry) electrodes capable of delivering an output in the range of mV, an instrumentation amplifier, a band-pass filter, and an amplifier.

For the system architecture design and hardware implementation consider that:

- OPAMP: TL084, MCP6004, LM324(?). If other is selected, check with the teaching staff about component availability.

- Instrumentation Amplifier: INA126, INA128, AD620, AD627 depending on the availability in the Lab. Due to limited component availability, not all the groups will use the same instrumentation amplifier model.
- Analog filters: cascade of second-order sections and first-order stage

This list is not complete. Please complete it before starting the design phase of the project.

2.2 - Microcontroller Unit (MCU)

The Raspberry Pi Pico 2 W is based on the RP2350 microcontroller featuring dual-core Arm Cortex-M33 processors (and optional RISC-V Hazard3 cores) running at up to 150 MHz. It offers 520 KB of SRAM onboard and comes with 4 MB of QSPI flash memory. The board integrates 2.4 GHz Wi-Fi (802.11n) and Bluetooth 5.2, making it well-suited for wireless IoT applications.

Peripherals include I²C, SPI, UART, PWM, ADC channels and 12 PIO state machines, enabling custom I/O and signal processing. Sleep and low-power modes are supported, with wake-up via GPIO, timer, or USB, which makes the Pico 2 W suitable for energy-sensitive embedded applications.

Note: the recommended MCU is the ESP32 and the available development board is the NodeMCU from Joy-IT. However, if a group wants to use other MCU (for example, Raspberry Pico Wi-Fi), contact the teaching staff.

2.3 - PMU

The power management unit (PMU) is responsible for delivering all required supply voltages to the system, ensuring that each supply is regulated to minimize ripple and maintain stability. This unit also manages energy storage components and supports multiple energy sources, such as various energy harvesters, if needed.

The system should be designed to operate with a maximum input supply of 9V, for example from a battery. To generate a stable 5V supply, the LM7805 linear regulator is recommended. Other DC-DC converter options are available, linear regulators are typically also used when very low ripple levels are required.

2.4 - Software and communications

For the software part, the solution should include:

- the firmware running in the microcontroller; (clearly include in your report, all the algorithms, program flow, and digital signal processing functions that are implemented).
- a software companion running in the computer, able to communicate with the microcontroller software via a serial USB connection;
- implementation of a GUI (see simplified example in figure 3)
- capability to send data wirelessly via WiFi to a computer or server.

This list is not complete. Please complete it before starting the design phase of the project.

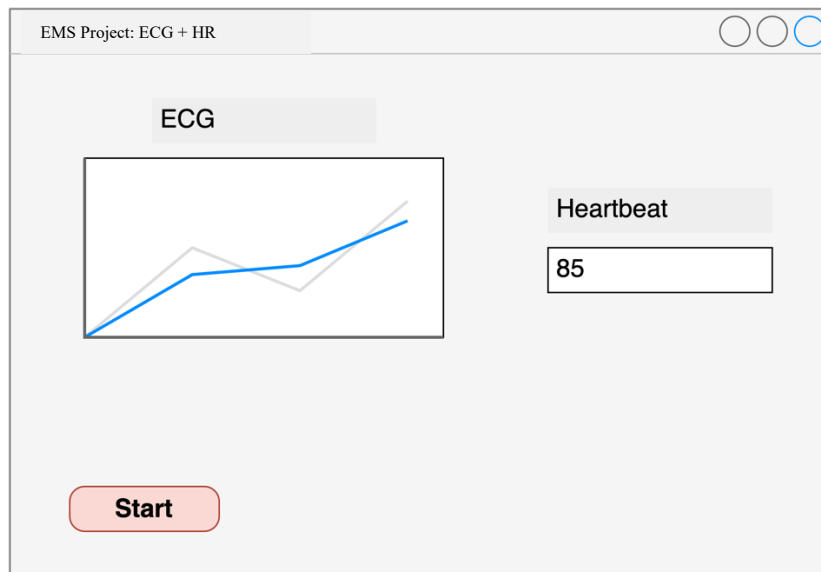


Figure 3 – Simplified GUI mockup. Can be improved.

2.4.1 MCU Firmware

The firmware for the RPI PICO 2 microcontroller (MCU) is a critical component, responsible processing the signal from the ADC, managing communication with the GUI and server, and optional sound effects.

Key Requirements:

- Sleep Mode and Wake-Up: The MCU should enter a low-power sleep mode when idle, waking up by a trigger event. This will require configuring the RPI PICO 2's wake-up sources based on either an external interrupt or internal timer.
- Programming Paradigms:
 - Basic implementations may use sequential programming for handling tasks.
 - Alternatively, more advanced implementations can incorporate interrupts to handle wake-ups and asynchronous tasks.
 - Alternatively, in an even more advanced implementation, one can use RPI PICO 2 RTOS capabilities, dividing tasks into separate threads (e.g., for SAD, sound effects, and data transmission), with task prioritization for efficient resource management.
- Communication Support:
 - USB/Serial Communication: The firmware must support serial communication to connect with the Python GUI application running on the PC. This connection will allow real-time monitoring and control from the GUI.
 - Wi-Fi and MQTT Protocol: The firmware should also establish a Wi-Fi connection to the MQTT broker, sending detected voice activity and command data to the server for storage and visualization.

Suggestion:

- Ensure modular code design to allow easy updates and debugging.

2.4.2 GUI Application on the PC

The GUI application provides an interface for interacting with the RPI PICO 2, monitoring the system's status, and adjusting configurations in real-time.

Key Requirements and optional:

- Serial Communication with RPI PICO 2: The GUI should connect via USB/Serial to the RPI PICO 2, receiving data on SAD status, and other system metrics.
- Real-Time Display: Display real-time data, such as sound level, with options for viewing command history.
- (optional) Control and Configuration Panel: Provide options for adjusting parameters, directly from the GUI to optimize performance based on environmental conditions.
- (optional) Data Logging and Export: Allow data to be logged and optionally exported for further analysis.

Suggestions:

- Develop the GUI in Python using libraries such as Tkinter for cross-platform compatibility.
- Implement a responsive layout and ensure that serial connection errors are handled gracefully.
- Consider additional features, such as alert notifications or command response logs, to enhance user interaction and control.

2.4.3 Server Application for Data Management and Visualization

The server-side application is responsible for storing, managing, and visualizing data from the RPI PICO 2. This includes MQTT messaging, data storage in InfluxDB, and visualization with Grafana.

Key Requirements:

- MQTT Broker: Use Mosquitto as the MQTT broker to facilitate communication between the RPI PICO 2 and the server. The RPI PICO 2 should publish data to designated topics, and the GUI may also subscribe to certain topics if needed.
- Node-Red may be used.
- Data Storage: Use InfluxDB to store time-series data for voice activity events, recognized commands, and other relevant metrics sent from the RPI PICO 2.
- Data Visualization with Grafana: Configure Grafana to visualize data from InfluxDB, displaying time-series charts for voice activity, command frequency, and any additional metrics. Set up dashboards to show real-time data as well as historical trends.

Suggestions:

- To simplify setup, we have prepared a Docker environment that includes pre-configured Mosquitto, InfluxDB, and Grafana services. Students are encouraged to use Docker for easy deployment.
- Ensure that data retention policies in InfluxDB are configured properly to avoid excessive storage use while maintaining relevant historical data.
- In Grafana, consider setting up alerts based on specific conditions for additional insights.

3- Project workflow Considerations

When planning the project workflow and timeline, consider the following topics:

- Project Start: consolidate requirements
- AFE hardware design, simulation, and implementation
 - Electrode input network interface
 - Instrumentation amplifier and any other amplifier
 - Analog filter
 - AFE simulation, implementation, and testing
- MCU software design and implementation
 - Signal processing (e.g., digital filter design)
 - Firmware design, implementation, and testing
- Computer companion script
 - Text interface
 - Simple GUI interface
- Wireless communication
 - Low priority
 - WiFi connectivity (for example, using an MQTT protocol messaging mechanism),
- Report and discussions (evaluation)

4- Report and Grading

The report should be complete with a detailed description of the design, simulation, implementation, and testing of the system. All netlists, python scripts, notebooks, excel files, and other files must be added to the submission pack.

Evaluation will be based on the quality of the design and implementation, achieved results, report quality. Table below describes the main criteriums for work evaluation. The knowledge and level of participation demonstrated by each group element during the final discussion will be considered in the final project classification grade.

Note: the project report shall detail the effective participation and work done by each element of the group.

Table 1 – Group Work and Report Grading Criteria

Criteria	Option A (up to 17)	Option B (up to 20)	Description
Core Completion	0 to 14	0 to 14	Complete implementation of the analog front-end with instrumentation amplifier, analog filtering, and MCU firmware for ECG acquisition, digital filtering, QRS/heartbeat detection, and GUI + IoT visualization. The filtering must target the ECG bandwidth ($\approx 0.5\text{--}40\text{ Hz}$) and support accurate heartbeat detection.
Advanced Signal Processing / Biomedical Analysis (only Option B)	-	0 to 3	Implementation of an extended DSP or system feature, such as real-time FFT or spectrum display, adaptive or digital notch filtering, multi-tasking/RTOS-based processing, or another technically creative enhancement.
Innovation (Extra Feature)	0 to 1	0 to 1	Creative additions or enhancements beyond the core requirements, such as adaptive filtering, additional commands, or custom feedback features in the GUI.
Report Quality	0 to 1	0 to 1	Clarity and depth in the report, including technical insights, individual contributions, challenges faced, and explanations of any innovative elements.
Overall Design and Functionality	0 to 1	0 to 1	Smooth integration of hardware and software components, functionality and usability of the GUI, and quality of data visualization in InfluxDB/Grafana.

Core completion and report submission are mandatory.

The project final grade assigned to each student will be composed of:

$$\text{Student Project Total Grade} = 75\% \times (\text{Group Work and Report Grade}) + 25\% \times (\text{Individual Achievement})$$

- Group Work and Report Grade (75%): reflects the overall quality of the project implementation, design, documentation, and demonstration, as described in Table 1.
- Individual Achievement (25%): based on each student's technical understanding, oral discussion performance, and demonstrated participation in the project.

References

Check Moodle page.